

Erdős Problem #374

EP-374

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

For any $m \in \mathbb{N}$, let $F(m)$ be the minimal $k \geq 2$ (if it exists) such that there are $a_1 < \dots < a_k = m$ with $a_1! \dots a_k!$ a square. Let $D_k = \{m : F(m) = k\}$. What is the order of growth of $|D_k \cap \{1, \dots, n\}|$ for $3 \leq k \leq 6$? For example, is it true that $|D_6 \cap \{1, \dots, n\}| \gg n$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>